

Timebox	4-5 hours. Use of AI tools is permitted; please state briefly where they were used.
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Objective

Build a small program or service that turns natural-language requirements into an intermediate semantic representation and uses that representation to evaluate requests. We are looking for clear modelling, careful reasoning and readable engineering - not a large system or polished UI.

Task

- Read the three requirement statements below and create an intermediate semantic representation.
- Evaluate each request against that representation and the evidence pack.
- Return one of four outcomes for each request: Approve, Reject, Revise, or Escalate.
- Explain which requirement, evidence condition or uncertainty drove each result.

This is not a flat JSON extraction exercise. Your representation should preserve enough meaning to support consistent downstream decision-making.

Decision meanings

Decision	Meaning
Approve	The request satisfies the relevant requirements and supporting evidence.
Reject	The request violates a non-negotiable condition.
Revise	The request is not acceptable as submitted, but could become acceptable through a bounded change, such as reducing scope, lowering usage, or providing an additional allowed evidence item.
Escalate	The request cannot be decided safely and automatically because required evidence is missing, incomplete, stale, conflicting, or the operating condition cannot be verified.

Requirement statements

ID	Requirement
R1	External partners may request aggregate analytical outputs from the protected dataset only if no individual-level information is disclosed and a valid approval record exists before execution.
R2	Any processing involving protected data must run in the approved local execution environment. If the execution environment cannot be verified, the request must not be automatically approved.
R3	Trusted internal teams may request a temporary increase above the standard usage limit, but only where a specific usage-adjustment approval exists. Disclosure restrictions and pre-execution approval conditions cannot be relaxed. If approval evidence is unclear, stale or conflicting, the case should be escalated rather than assumed safe.

Candidate input pack

Request pack

ID	Requester / trust	Action / output	Environment / usage	Evidence
R1	external_partner; external	aggregate_analysis; aggregate_counts; protected_dataset	local_approved_env; standard	E1, E2
R2	external_partner; external	row_level_export; individual_records; protected_dataset	local_approved_env; standard	E1, E2
R3	internal_team; trusted_internal	aggregate_analysis; aggregate_counts; protected_dataset	local_approved_env; above_standard_limit	E1, E2
R4	external_partner; external	aggregate_analysis; aggregate_counts; protected_dataset	unverified_remote_env; standard	E1
R5	internal_team; trusted_internal	aggregate_analysis; aggregate_counts; protected_dataset	local_approved_env; above_standard_limit	E2, E3, E4

Evidence pack

ID	Type	Details
E1	approval_record	status=valid; timing=before_execution; reviewer=designated_reviewer; timestamp_state=current
E2	execution_environment_attestation	subject=local_approved_env; status=verified; attestation_state=valid
E3	usage_limit_adjustment	status=approved; scope=trusted_internal_only; adjustment_type=above_standard_limit; reviewer=designated_reviewer; timestamp_state=current
E4	approval_record	status=conflicting; timing=before_execution; reviewer_state=one_valid_one_revoked; timestamp_state=conflicting

What to submit

Deliverable	Expectation
Source code	Runnable implementation, with clear structure and minimal setup.
README	How to run, dependencies, and input/output format.
Design note	Maximum one page: explain the intermediate representation, why it is more useful than flat extraction, how decisions are made, where escalation occurs, and what you would improve next.
Outputs for R1-R5	Machine-readable results for all five requests. Suggested fields: request_id, decision, rationale, requirements_applied, evidence_used, missing_or_conflicting_evidence, assumptions, unresolved_uncertainty.
Optional tests	Automated tests are welcome but not mandatory. Meaningful edge cases matter more than volume.

Minimum expectations and evaluation

Minimum expectations	Evaluation focus
Define an intermediate representation; do not hardcode each request. Distinguish non-negotiable constraints from revisable conditions. Treat missing or conflicting evidence explicitly. Produce bounded explanations.	Semantic modelling; decision quality; uncertainty handling; engineering quality; clarity of reasoning and assumptions.

Note: We prefer honest partial work with clear reasoning over polished but opaque submissions.